SUSTAINABILITY

From Silicon to System: Embedding Power Efficiency as a Core Design Principle for Scalable AI and Compute Infrastructure

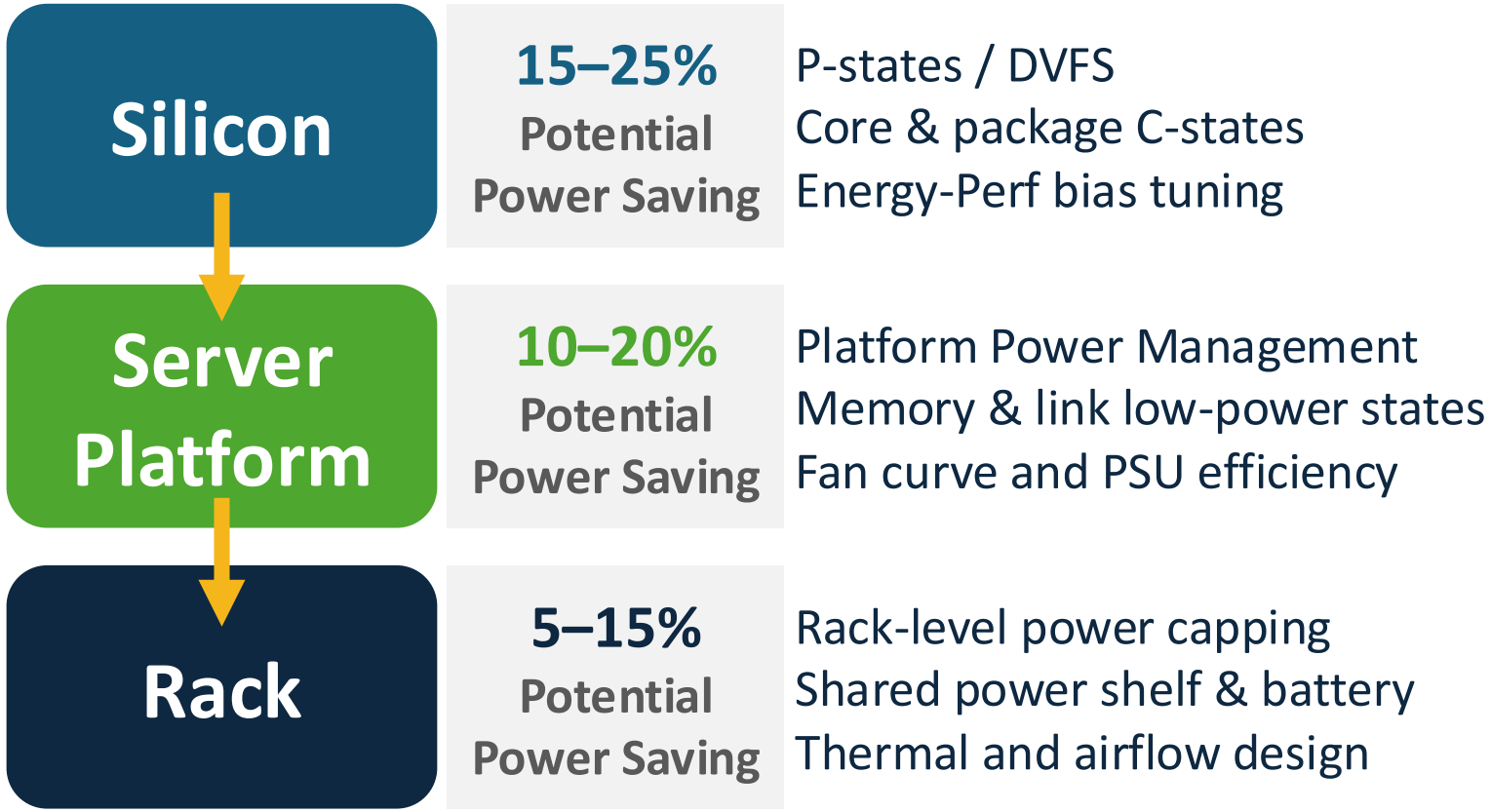
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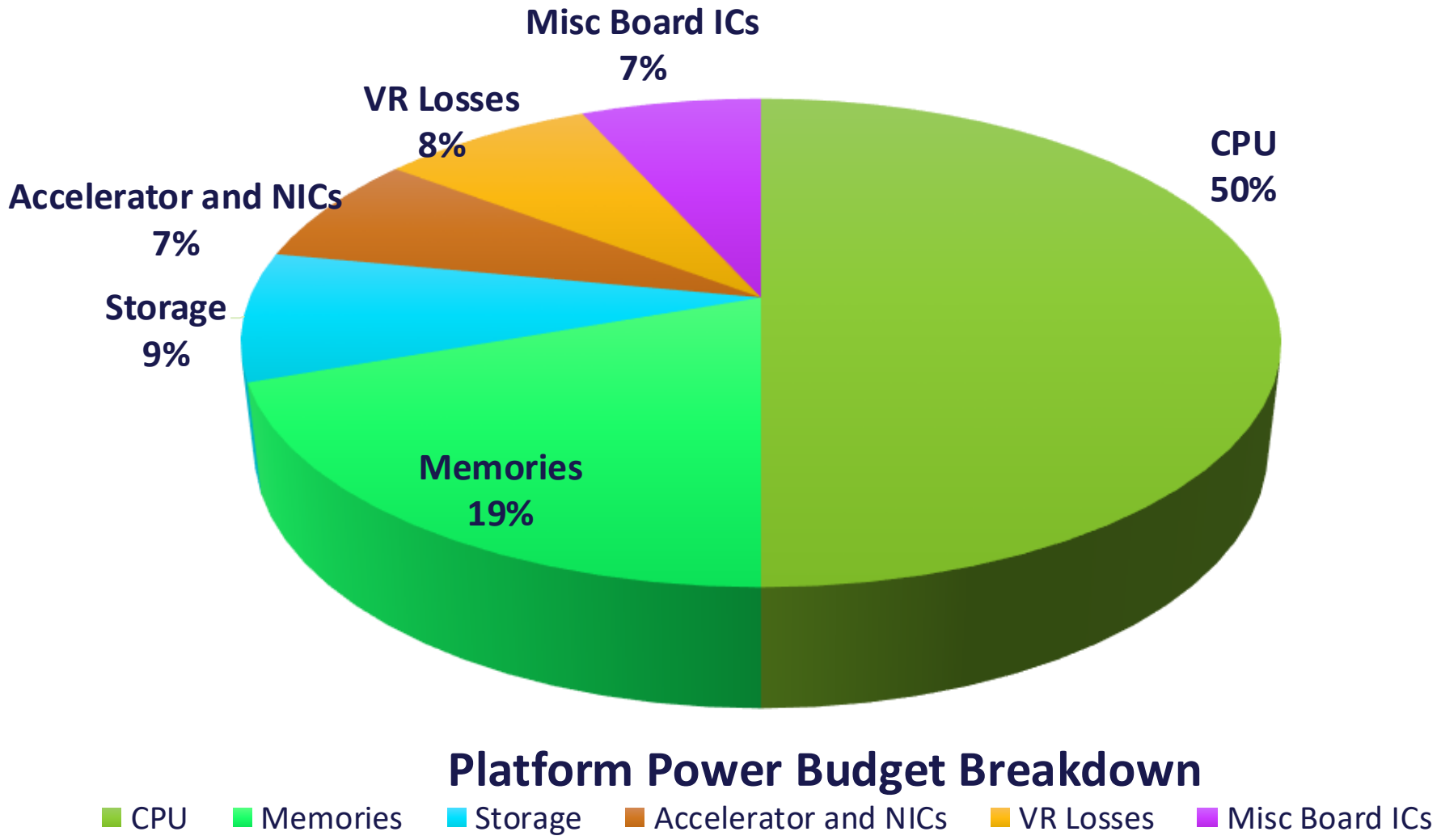
Power Efficiency as a Core Design Principle

- Power savings compound from silicon up to the rack
- No single layer reaches the target alone
- Each layer needs its own owner and budget



Power has become the most fundamental constraint in AI and Compute Data Centers build out

Example of Where the Power Budget Goes

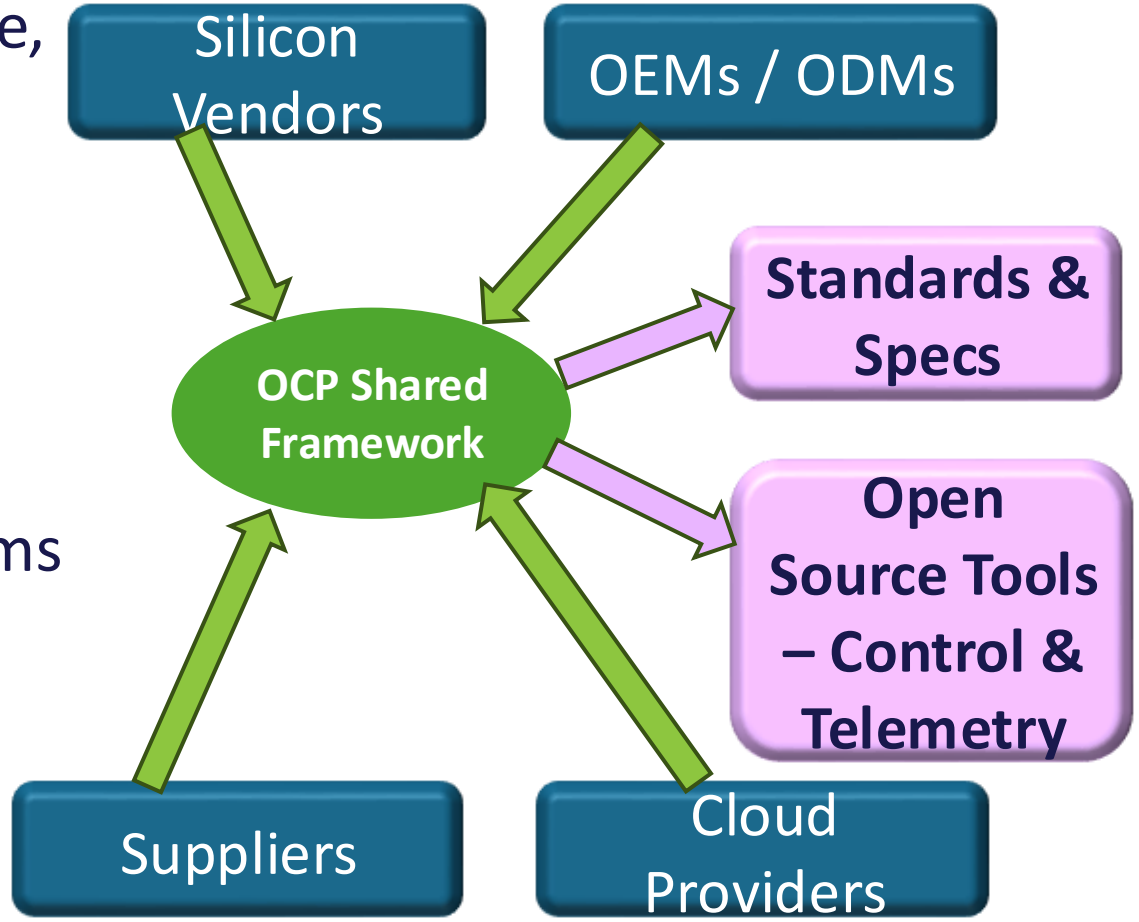


Why This Matters to OCP

- Power reduction across CPU, Memory, Storage, Accelerators needs to be aligned.
- Carbon neutral goals, many targeting 2030
- AI demand outpaces facility efficiency alone
- Every watt saved compounds downstream
- Shared metrics move the curve, not single firms



Aligned action compounds impact



Building A Structured Framework

Power-Saving Feature	Status	Details (BIOS/BMC/OS knobs, defaults)	Area	Notes / Rationale
P-States (DVFS)	Enabled	BIOS default profile; OS governor = balanced	HW/FW	Primary lever under partial load
Core C-States (C1E/C6)	Enabled	C6 enabled; wake latency within SLA	FW/OS	Largest idle-power contributor
Package C-States (PCx)	Enabled	PC6 enabled at BIOS default	FW	Gates LLC/uncore at low utilization
Turbo Boost	Enabled	Capped by platform power limit	HW/FW	Balanced against power ceiling
Energy-Perf Bias (EPB)	Disabled	Set to Performance for this SKU	OS	Rationale and exception path required

Example of CPU Power-Saving Feature Checklist

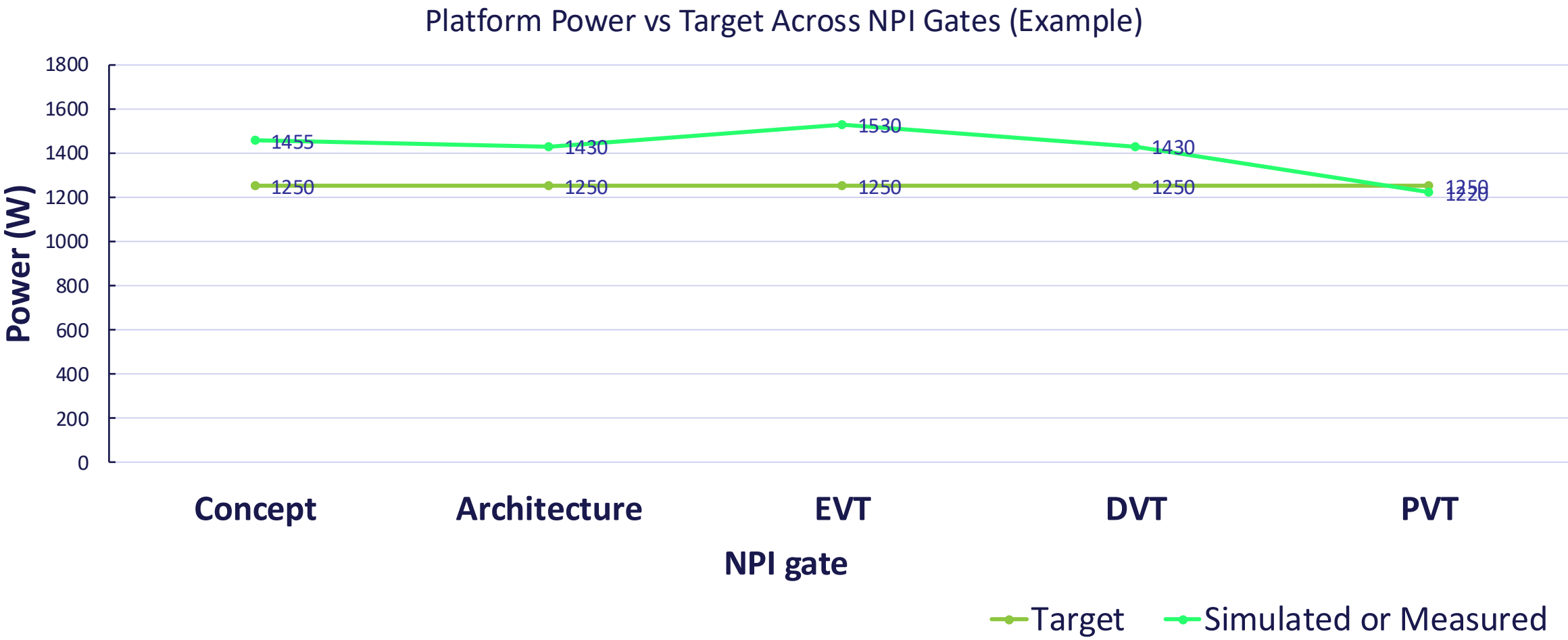
Build a shared set of requirements for power saving capabilities across the industry for the CPU, Memories, Storage, Accelerators, etc.

Extend to rack-level policies, requirements that can create more granular and standardized datacenter–level capabilities.

The capabilities need to be supported under a common framework of open-source tools, to maximize power efficiency across deployment classes.

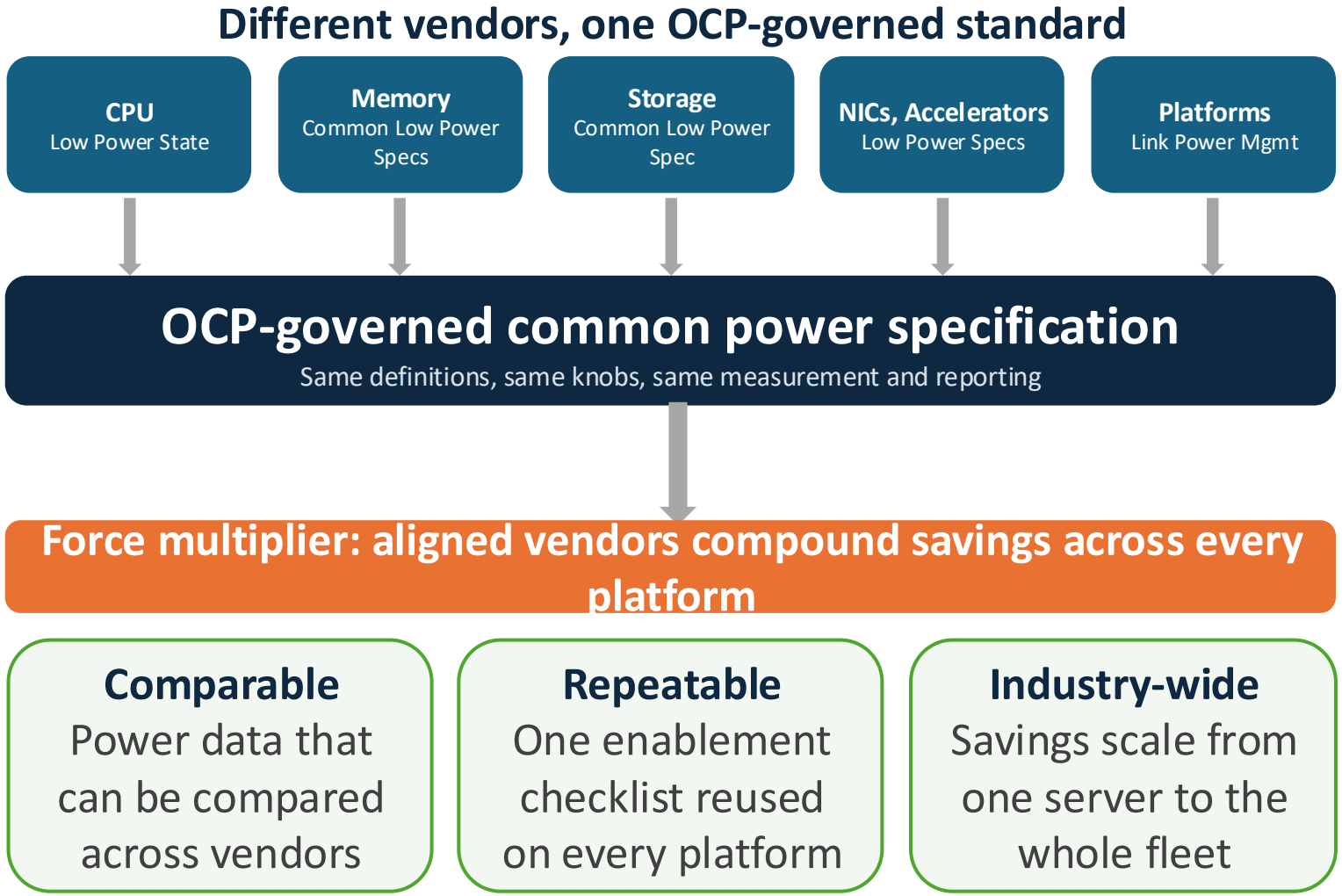
Inviting the industry to expand this model to build a common ecosystem, from components to fleet management.

Example: Closing the Power Gap to Target



Force Multiplier: One OCP Standard

- OCP governs one common power specification
- CPU, memory, storage, accelerator, NIC & platform vendors complies to it.
- Shared knobs and measurement remove per-vendor rework
- Alignment turns single-box gains into industry-wide savings



Call to Action

- **Problem to solve:** Power efficiency is not designed in up front, enabled and validated late. Build a common silicon-to-rack power specification for CPU, memory, storage, accelerators, NICs, and PCIe.
- **Get involved:** Join the OCP server and sustainability workstream calls to co-develop it
- **Timeline:** Draft spec and CPU feature checklist for review in two quarters; remaining building blocks next
- **Adopt the framework** in your next NPI cycle, with measured-vs-target power at every design gate
- More information: https://www.opencompute.org/projects

Thank You!

